ticking of the sliding clock becomes slower and slower as it moves faster and faster.2

To get a sense of scale, note that the photon traverses one round-trip in about a billionth of a second. For the clock to be able to travel an appreciable distance during the time for one tick it must therefore be traveling enormously quickly—that is, some significant fraction of the speed of light. If it is traveling at more commonplace speeds like 10 miles per hour, the distance it can move to the right before one tick is completed is minuscule—just about 15 billionths of a foot. The extra distance that the sliding photon must travel is tiny and it has a correspondingly tiny effect on the rate of ticking of the moving clock. And again, by the principle of relativity, this is true for all clocks—that is, for time itself. This is why beings such as ourselves who travel relative to one another at such slow speeds are generally unaware of the distortions in the passage of time. The effects, although present to be sure, are incredibly small. If, on the other hand, we were able to grab hold of the sliding clock and move with it at, say, three-quarters the speed of light, the equations of special relativity can be used to show that stationary observers would see our moving clock ticking at just about two-thirds the rate of their own. A significant effect, indeed.

Life on the Run

We have seen that the constancy of the speed of light implies that a moving light clock ticks more slowly than a stationary light clock. And by the principle of relativity, this must be true not only for light clocks but also for any clock—it must be true of time itself. Time elapses more slowly for an individual in motion than it does for a stationary individual. If the fairly simple reasoning that has led us to this conclusion is correct, then, for instance, shouldn’t one be able to live longer by being in motion rather than staying stationary? After all, if time elapses more slowly for an individual in motion than for an individual at rest, then this disparity should apply not just to time as measured by watches but also to time as measured by heartbeats and the decay of body parts. This is the case, as has been directly confirmed—not with the life expectancy of humans, but with certain particles from the microworld: muons. There is one important catch, however, that prevents us from proclaiming a newfound fountain of youth.